Variational Autoencoder (VAE) for Satellite Image Generation – Executive Summary

Overview

The given experiment investigates how a Variational Autoencoder (VAE) can be learned to generate meaningful representations of EuroSAT dataset satellite images. The model was trained to encode satellite images into a latent space, which is compressed, and then reconstructed to create the satellite images with the help of a decoder network. The model learns this compressed representation, which contains spatial patterns and textures that exist in land-use images in the form of rivers, forests, and highways.

Training Performance

The training loss curve shows that the model steadily decreased the loss across several epochs, which shows that the model has been slowly improving its reconstruction ability. This indicates that the encoder and decoder networks were able to learn meaningful latent codes of the satellite imagery.

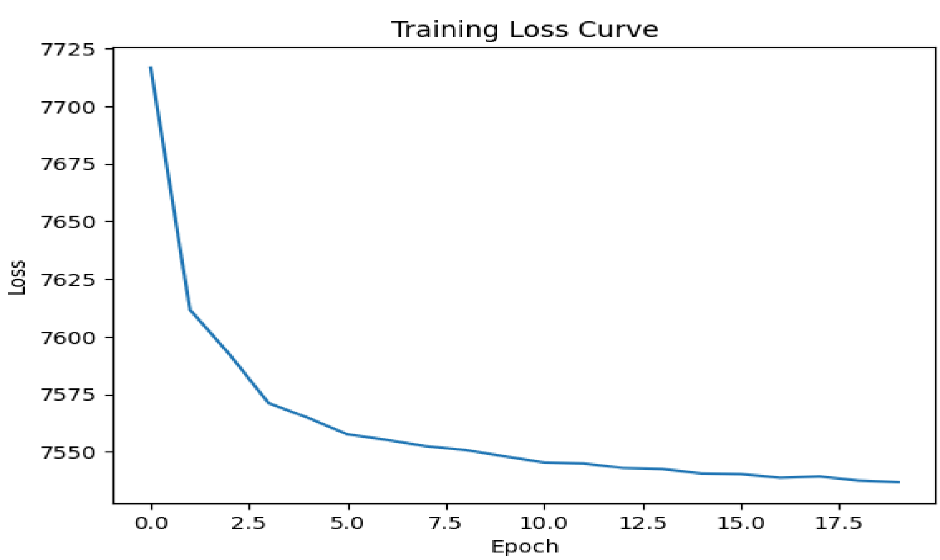

Figure 1: Training Loss Curve

Generated Image Samples

The obtained samples are used to show that the trained VAE is able to generate new satellite-like textures. Even though the images are a bit blurry, as is normal in VAE models, because it is a probabilistic model, it does not blur the spatial features like the pattern of the vegetation and surface variations.

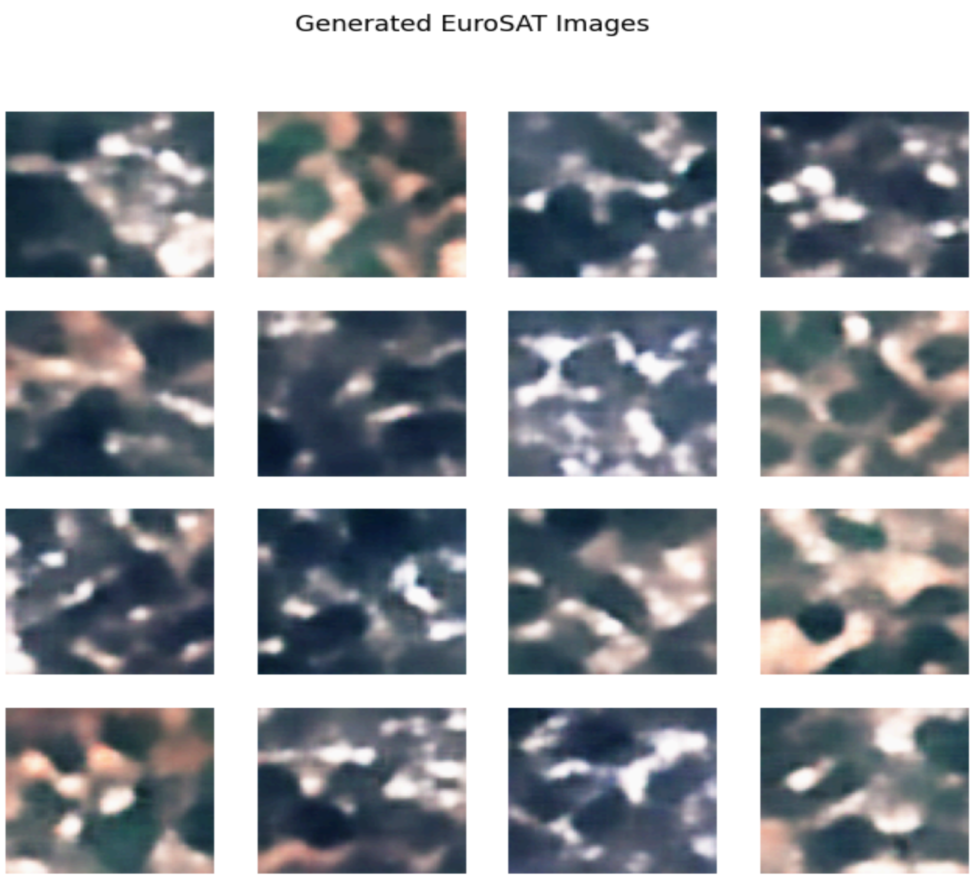

Figure 2: Generated EuroSAT Images

Latent Space Interpolation

Latent space interpolation shows that the learned representation is continuous. Through progressive interpolation of two latent vectors, there is a smooth change in patterns between the decoded image patterns. This implies that the latent space has similar structures of land-use which are close to one another.

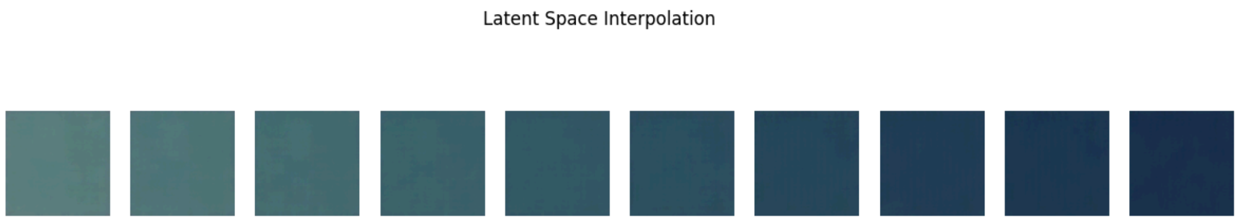

Figure 3: Latent Space Interpolation

Land-Use Manipulation Experiment (Task 4)

In order to experiment with semantic manipulation in the latent space, the mean latent representations of Forest and Highway classes were calculated. The distance between these vectors constituted a semantic direction known as the Paving Direction. This then was used to apply to the latent representation of a River image and the image then was decoded back to the image space.

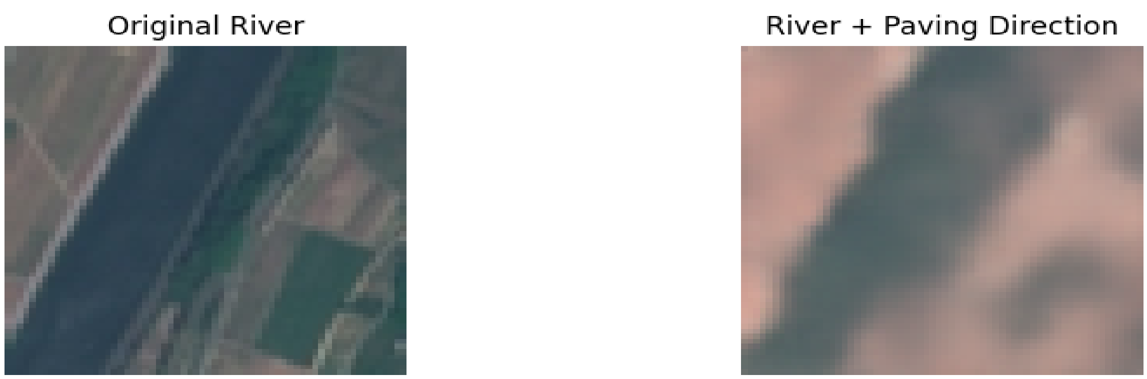

Figure 4: River Image Before and After Applying the Paving Direction

Conclusion

This experiment proves that the VAE learned meaningful land-use features representations in satellite imagery. According to the latent space arithmetic experiment, it is implied that the model represents semantic directions reflecting surface properties of vegetation and paved land. Despite the fact that the produced outputs are to a certain extent blurred by the probabilistic form of VAEs, the findings show that the model has managed to include high-level spatial patterns and associations between different land-use categories.